

Edge API & Database Architecture Specification

apps/api: 31 Prisma Relational Models, Cloudflare Workers / Edge Runtime & SSE Engine

APPS/API & DB

Project: RVCC Enterprise Monorepo	Version: 2.0.0 (Production)	Architecture: Next.js 16 + Edge + Prisma	Status: Verified & Active
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1. Architecture Overview & Runtime Stack

`apps/api` is the high-throughput Edge Backend API powering the entire RVCC monorepo. Engineered on the **Cloudflare Workers / Node Edge Runtime** using **Prisma ORM** with the `@prisma/adapters-pg` connection driver, it delivers sub-millisecond query execution, real-time Server-Sent Events (SSE) streaming, and cryptographically secure authentication.

Backend Technology Stack • Runtime: Cloudflare Workers / Node.js 22+ Edge Runtime. • Database ORM: Prisma Client v5.22+ with custom soft-delete extensions. • Database Engine: PostgreSQL 16 (Enterprise Schema). • Real-Time: Server-Sent Events (SSE) streaming bus.	Security & Reliability • Authentication: Session JWTs + SHA-256 OTP challenge tokens. • Auditing: Comprehensive unified AuditLog for all state mutations. • Soft Deletes: Zero data loss via automated <code>deletedAt</code> filtering. • Testing: 100% passing Vitest test suites.
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2. Complete Prisma Database Entity Model (31 Models)

The relational database schema encapsulates the entire business logic across 6 distinct domain areas:

Domain Area	Entity Models	Key Relationships & Purpose
1. Staff & RBAC	AdminUser , Role , Permission , RolePermission , AdminSession , AdminOtp , AdminLoginHistory	Granular access control, session token storage, password reset OTP challenges, and chronological login history.
2. Vendor & Auth	VendorUser , VendorSession , VendorOtp , VendorLoginHistory , SupplierRegistration	Supplier accounts, passwordless OTP authentication, session lifecycle, and registration dossiers.
3. Supplier Dossier	CompanyProfile , SupplierContact , SupplierAddress , BusinessClassification , BankAccount , QuestionnaireAnswer , RegistrationAttachment	Complete 7-part supplier qualification profile including legal, banking, contact, and compliance documentation.
4. Sourcing & Bidding	Requirement , RequirementInvite , Quote , ExchangeRate , Industry	Tender requirements, itemized BOQ, targeted supplier invites, multi-currency vendor quotes, and live FX rates.
5. Procurement PR	PurchaseRequest , PurchaseRequestItem , PurchaseRequestAttachment	Departmental purchase requisitions, itemized breakdowns, budget approvals, and sourcing links.
6. Content & Audit	Notification , AuditLog , JobPosting , JobApplication	Real-time user notifications, immutable audit log ledger with JSON state diffs, careers CMS and candidate applications.

3. Prisma Client Soft-Delete Extension Architecture

All core tables implement soft-deletion to ensure audit compliance and prevent catastrophic data loss:

```
// apps/api/src/db.ts
export const prisma = new PrismaClient().$extends({
  query: {
    $allModels: {
      async delete({ model, args }) {
        return (prisma[model] as any).update({
          ...args,
          data: { deletedAt: new Date() }
        });
      },
      async findMany({ model, args }) {
        args.where = { deletedAt: null, ...args.where };
        return (prisma[model] as any).findMany(args);
      }
    }
  }
});
```

4. Real-Time SSE Live Bidding Bus Architecture

The Server-Sent Events engine (`modules/bidding/live-bids.ts`) broadcasts price ranking updates to all connected cockpits with zero overhead:

- **Broadcast on Quote Submission:** When any vendor submits a revised quote to `/api/requirements/[id]/quote`, the API normalizes the price to SAR, updates the database in a transaction, recalculates the L1 ranking array, and immediately broadcasts the updated payload to all open SSE connections.
- **Payload Anonymization:** The SSE broadcaster generates two separate payloads:
 - **Admin Payload:** Contains full vendor identity and quote details for evaluation.
 - **Vendor Payload:** Cryptographically masks all competitor names (e.g. `"Competitor A"`, `"Competitor B"`) while publishing the exact L1 market price and rank positions.

5. Daily FX Synchronization Engine

To guarantee accurate multi-currency rankings, an automated cron service runs every 24 hours:

- Fetches official daily exchange rates for `USD`, `EUR`, `GBP`, `AED`, `QAR` against base `SAR`.
- Updates the `ExchangeRate` table in PostgreSQL.
- Caches rates in memory to guarantee sub-millisecond conversion calculations during high-frequency bidding events.

6. REST API Endpoint Registry Summary

Endpoint Group	Methods	Key Operations
/api/auth/*	POST	Admin & Vendor login, logout, OTP challenge generation, password reset.
/api/requirements/*	GET, POST, PUT	Create, publish, list, inspect RFQs, submit quotes, award contracts.
/api/requirements/:id/live	GET (SSE)	Server-Sent Events streaming feed for real-time live bidding cockpits.
/api/registrations/*	GET, POST, PUT	Vendor self-registration train, document review, approval/rejection.
/api/procurement/*	GET, POST, PUT	Purchase requisition creation, manager approvals, sourcing conversion.
/api/staff/*	GET, POST, PUT, DELETE	Staff account administration, role assignment, permission governance.